

Course Syllabus

San José State University

Department of Applied Data Science

DATA 260, Agentic Artificial Intelligence and Distributed Systems

Course and Contact Information

Instructor:

Simon Shim, Ph.D.

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Office Hours: Monday, 1:00pm – 3:00pm

Class Days/Time: Monday, 3:00pm – 5:45pm

Classroom: ENG 343

Prerequisite(s): [DATA 226](#) 

Prerequisites: (https://catalog.sjsu.edu/preview_program.php?catoid=23&poid=18795&returnto=8470#tt8209) (with a “C” or better grade).

Course Format

This course adopts a in-person classes. Students must ensure that the laptop is in a good condition to be able to take the quizzes and exams on Canvas, which may include Respondus monitoring using the laptop’s camera. Exams missed or jeopardized due to technical issues with the laptop may not be re-administered.

Course Web Page and MYSJSU Messaging

Course materials such as syllabus, handouts, notes, assignment instructions, etc. can be found on [Canvas Learning Management System course login website](#) at . You are responsible for promptly checking with the messaging system through Canvas, regular email and [MySJSU](#) at to learn of any updates every day.

Course Description

Learn agentic artificial intelligence in the context of distributed systems. Cover distributed design, messaging systems, pub/sub messaging, cloud-based architectures, and REST services while integrating agent capabilities such as tool/function calling, retrieval-augmented generation with vector stores, short-/long-term memory, planner-executor and ReAct loops.

Course Goals

Upon successful completion of this course, students are expected to:

Description

CLO 1 Understand the architectures of distributed systems

CLO 2 Design and implement distributed applications and services

CLO 3 Develop agentic AI systems

CLO 4 Design and evaluate multi-agent and distributed AI architectures

CLO 5 Apply critical thinking in agentic AI solutions

Course Learning Outcome

1. Demonstrate an understanding of distributed systems architectures [PLO 2]
2. Demonstrate an understanding of distributed systems architectures, methodologies, frameworks, and tools [PLO 2]
3. Demonstrate leadership and the ability to participate in teamwork [PLO 6]

Required Texts/Readings

It is not required to buy a textbook for this course.

Other technology requirements equipment/material

Python (Scikit-learn, TensorFlow, PyTorch, Matplotlib, OpenCV,) is required for this course.

Other Readings

Roy Fiedling, [Architecture Styles and Design of Network based Software](https://www.ics.uci.edu/~fiedling/pubs/dissertation/fiedling_dissertation.pdf) 
(https://www.ics.uci.edu/~fiedling/pubs/dissertation/fiedling_dissertation.pdf)

Simon Shim, [The CAP Theorem's Growing Impact](https://www.computer.org/cms/Computer.org/ComputingNow/homepage/2012/0312/W_CO_TheCAPTheorem) 
(https://www.computer.org/cms/Computer.org/ComputingNow/homepage/2012/0312/W_CO_TheCAPTheorem)

Research papers may be assigned during the course.

Canvas

- Canvas will be the primary learning management system for this course. It will be used to:
 - Distribute lecture notes, readings, examples, and other course materials.
 - Post announcements, schedules, assignment instructions, and due dates.
 - Facilitate online discussions and other course-related activities.
 - Collect assignments, projects, reports, and other required submissions.
 - Provide grades, feedback, and grading rubrics.
 - Support academic integrity and plagiarism detection, when appropriate.

Unless otherwise stated by the instructor, assignments must be submitted through Canvas by the specified deadline in order to be considered for grading.

Course Requirements and Assignments

Students are expected to actively participate in class discussions, demonstrations, exercises, and collaborative activities. Active participation can contribute significantly to understanding distributed

systems, agentic AI, and their practical applications.

Students should come to class prepared by completing assigned readings, reviewing course materials, and completing any required preparation before class. Students are expected to remain engaged during class and avoid activities that interfere with their own learning or that of others.

Course assignments may include individual programming assignments, laboratory exercises, technical analyses, agentic AI and distributed systems applications, presentations, and team-based projects. Detailed instructions, grading criteria, and due dates will be provided through Canvas.

For project-based assignments, students may be expected to design, implement, evaluate, and document distributed and/or agentic AI systems using technologies and frameworks introduced in the course. Team projects may also require students to demonstrate effective collaboration, communication, and division of responsibilities.

SJSU courses are designed with the expectation that, for each unit of credit, students will spend a minimum of 45 hours over the duration of the course on instruction, preparation, studying, course-related activities, assignments, projects, and other academic work. For a three-unit course, this corresponds to approximately nine hours of course-related work per week, including scheduled class time.

Additional information regarding student workload expectations is provided in SJSU University Policy S12-3.

Determination of Grades

The percentage weight assigned to class assignments is listed below. Detailed grading rubrics for the significant paper assignment, team project, and exact due dates for each assignment will be posted on Canvas.

Students will have at least one week to complete each homework assignment.

Homework assignments	10%
Course Group Project	20%
Quizzes	10%
Labs	20%
Midterm Exam	15%
Final Exam (comprehensive)	25%

Descriptions of Assignments/Exams

Course Group Project: Based on the ideas gathered from the course, a group will work on a term-long group project related to the topics covered as part of the course. The project has deliverables throughout the semester. The quality and completeness of all the deliverables will be considered in grading the projects. All projects will be demonstrated in class. Each group member is expected to participate in every phase of the project. The final grade of each member will be proportional to his/her participation in the group, as assessed by the instructor. Each member should be able to answer questions regarding the project, present some part of the project demo, and participate in the system implementation and the writing of the technical reports. The term project will be graded on the basis of the following three components: a) project implementation, b) project report, c) project demonstration.

In-class quizzes: Students will be given in-class quizzes that will test their understanding and retention of presented material. All quizzes and exams are comprehensive, covering all topics completed till then. The worst quiz grade will not be counted towards the final in-class quiz grade of each student ("worst-one out policy").

Programming and other Homework assignments: As part of the homework, students will be provided instructions describing programming assignments and exercises related to database or data warehouse problems. Generally, there are many ways to solve a problem. As such, students are encouraged to try multiple ways and compete against their peers to achieve the best result.

Exams: Exams will comprise of multiple-choice questions and will be based on the individual assignments, project work, and all course material covered in class till that time. All quizzes and exams are comprehensive and will be recorded using the student's laptop and camera. No make-up exams will be administered even if

a student misses an exam (unless the student has a written pre-approval by the instructor for completely unavoidable reason such as illness or other personal emergencies and can provide documented evidence).

NOTE that University policy F69-24 states that "Students should attend all meetings of their classes, not only because they are responsible for material discussed therein, but because active participation is frequently essential to insure maximum benefit for all members of the class. Attendance per se shall not be used as a criterion for grading."

Final Examination or Evaluation

This course has a comprehensive final exam. The date is determined by the University's Final Examination Schedule

Grading Information (Required)

The final grades will be calculated based on the following (subject to change, based on the class performance):

<i>Grade</i>	<i>Percentage</i>
<i>A plus</i>	<i>96 to 100%</i>
<i>A</i>	<i>93 to 95%</i>
<i>A minus</i>	<i>90 to 92%</i>
<i>B plus</i>	<i>86 to 89 %</i>
<i>B</i>	<i>83 to 85%</i>
<i>B minus</i>	<i>80 to 82%</i>
<i>C plus</i>	<i>76 to 79%</i>
<i>C</i>	<i>73 to 75%</i>
<i>C minus</i>	<i>70 to 72%</i>
<i>D plus</i>	<i>66 to 69%</i>
<i>D</i>	<i>63 to 65%</i>
<i>D minus</i>	<i>60 to 62%</i>
<i>F</i>	<i>0 to 59%</i>

- The above may be scaled up or down based on class
- Please note from Canvas, the day and time when each assignment is Late assignments will incur incremental penalty of 20% each day they are late. Please do not ask for date extensions.
- Exceptions may be accommodated only if a student has serious and compelling reasons that can be proven by an independent authority (e.g. doctor's note if the student has been sick).

Classroom Protocol

Students are expected to arrive in time for class. During the class, they need to turn off cellphones unless directed otherwise by the instructor. Laptop/tablet/smart phone use is allowed only for activities related to the class. Please be considerate of your fellow students. Students may not record (audio or video) in this class except in accordance with ADA accommodations. Any recordings made in connection with a disability accommodation are for the student's personal academic use only and may not be distributed in any manner to any other individual.

University Policies

Per University Policy S16-9, university-wide policy information relevant to all courses, such as academic integrity, accommodations, etc. will be available on Office of Graduate and Undergraduate Programs' [Syllabus Information web page](#) at "

Technology Requirements

Students are required to have an electronic device (laptop, desktop, or tablet) with a camera and microphone. SJSU has a free [equipment loan program](#) available for students.

Classroom Protocol

- Classes will start on time, so do not be
- If a student misses a class, the student is responsible for finding out what's said/done in that class (such as new announcement, deadline change, etc.) and responding accordingly.
- Please make sure to turn off or mute the cell phone during
- Please do not perform irrelevant or distracting activities in

Students Are Not Allowed to Record or Share Materials outside Class

Students are prohibited from recording class activities (including class lectures, office hours, advising sessions, etc.), distributing class recordings, or posting class recordings. Materials created by the instructor for the course (syllabi, lectures and lecture notes, presentations, etc.) are copyrighted by the instructor. This university policy ([S12-7](#)) is in place to protect the privacy of students in the course, as well as to maintain academic integrity through reducing the instances of cheating. Students who record, distribute, or post these materials will be referred to the Student Conduct and Ethical Development office. Unauthorized recording may violate university and state law. It is the responsibility of students that require special accommodations or assistive technology due to a disability to notify the instructor.

Proctoring Software and Exams

Examinations in this course may be administered **electronically or on paper**, at the instructor's discretion.

For online examinations, the instructor may use proctoring tools such as **Respondus Monitor, LockDown Browser, and/or Zoom**. The specific examination format and proctoring method will be announced in advance.

For in-person examinations, the instructor may administer the exam using a **paper-based format** or another appropriate method.

If academic misconduct is suspected during a proctored examination, relevant proctoring records or recordings may be reviewed as part of a further investigation and may be used in accordance with university academic integrity policies. Proctoring software itself does not determine whether academic misconduct has occurred; it may identify irregularities that require additional review.

Students should contact the instructor as soon as possible if unexpected technical problems or unavoidable interruptions occur during an examination.

Academic Dishonesty

Technical Support for Canvas Email: [ecampus@sjsu.edu \(mailto:ecampus@sjsu.edu\)](mailto:ecampus@sjsu.edu) Phone: (408) 924-2337

Students who are suspected of cheating during an exam will be referred to the Student Conduct and Ethical Development office and depending on the severity of the conduct, will receive a zero on the assignment or a grade of F in the course. Grade Forgiveness does not apply to courses for which the original grade was the result of a finding of academic dishonesty.

University Policies

Per University Policy S16-9, university-wide policy information relevant to all courses, such as academic integrity, accommodations, etc. will be available on Office of Graduate and Undergraduate Programs' [Syllabus Information web page](#) at "

Department Policies

- Students who do not provide documentation of having satisfied the class prerequisite or co-requisite requirements (if any) by the second class meeting will be dropped from the class.
- All non-proctored report (or similarly sized) assignments in courses where some of the final grade depends on prose writing will be submitted to Turnitin.com.
- Major exams in this class will be video recorded to ensure academic The recordings will only be viewed if there is an issue to be addressed. Under no circumstances will the recordings be publicly released.

List the agenda for the semester including when and where the final exam will be held. Indicate the schedule is subject to change with fair notice and how the notice will be made available.

Course Schedule

The schedule (and related dates/readings/assignments) is tentative and subject to change with fair notice. In case of guest lectures, the syllabus will be updated accordingly. Any changes will be announced in due time in class and on the course's web site (Canvas). The students are obliged to consult the most updated and detailed version of the reading material and syllabus, which will be posted on Canvas.

Week	Date	Topics, Readings, Assignments, Deadlines	Assignment Handout Date
1	8/24	Course Introduction, JavaScript, AWS ECS,Docker, HTML5, CSS, Introduction to Agentic AI	Refresher HW
2.	08/31	HTTP, REST, FastAPI, Agentic Behavior, A2A interactions	
3	09/07	Labor Day	
4	09/14	FastAPI, Session Mgmt, Prompt engineering, LLM introduction .	Lab 1
5	09/21	Bootstrap, React, JSX, MySQL, Context Engineering	Quiz 1,
6	09/28	React Advanced, FastAPI Agents, MCP in Action, Harness	

7	10/05	NoSQL DB (MongoDB)/Deployment to Cloud, LangChain Memory, Long term memory, Langraph, state checkpointing	Quiz 2, Class Project
8	10/12	Midterm Exam	
9	10/19	JWT, claims, and authorization, ReAct, MRKL, & other single-agent design patterns, agent security, tool permissions, human approval, prompt injection, Reflexion-style self-correction	Lab 2; Lab 1 due, Quiz 3
10	10/26	Distributed systems, Message Queues (Kafka Partitions, Cluster, Topics), Jmeter, multi-agent communication & coordination, multi-agent solutions, Hermes Agent, Redis caching, caching for repeated AI queries	
11	11/02	Coding Agents, Coding Agent Patterns, agent runtime measurements: token/cost/latency/tool success, Planning and reflection patterns, Self-improving agents	Quiz 4
12	11/09	Streamlit, Evaluation: core metrics, human-in-the-loop, benchmarking, G-eval, AI Testing, full Observability + Tracing + Evaluation week	
13	11/16	DevOps, Domain-specific agentic solutions (dev agents, simulation, knowledge agents), production monitoring + CI evals + regression testing	Lab 2 due, Quiz 5
14	11/23	CAP/Web Architecture, auto-GPT, ML infrastructure and model drift, Domain-specific agentic solutions, LLM/agent drift + production failure analysis	
15	11/30	Project Presentation	
16	12/07	Project Demo	Class Project
Final Exam	12/11	Fri, December 11, 3:15 AM-5:15 PM	

